Flower Species Recognition

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Abstract—This paper presents a deep learning approach to flower species recognition using Transfer Learning with MobileNet pre-trained on ImageNet. The system classifies images into five flower categories: daisy, dandelion, rose, sunflower, and tulip. Working with a limited dataset of 4,317 images, we combine data augmentation with a two-phase training strategy : feature extraction followed by fine-tuning of the last layers of MobileNet. Our approach achieves 87.9% validation accuracy, showing that Transfer Learning is an efficient strategy for image classification tasks with limited data.

Index Terms—Deep Learning, Transfer Learning, Convolutional Neural Networks, Flower Recognition, MobileNet

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I. INTRODUCTION

Flower recognition is a fundamental problem in computer vision with real applications in botany, agriculture, and environmental monitoring. Identifying flower species from photos is a challenge because the same species can look very different depending on lighting, orientation, and scale, while different species can also look very similar to each other.

Deep learning, and in particular Convolutional Neural Networks (CNNs), have shown remarkable performance in image classification tasks. However, training a model from scratch requires large amounts of labeled data. In practice, datasets are often small (In our case 4317 files), so strategies like Transfer Learning, data augmentation, dropout, fine-tuning are needed to reach high accuracy.

In this paper, we present a flower classification system using the Flowers Recognition dataset from Kaggle, which contains five categories: daisy, dandelion, rose, sunflower, and tulip. We use MobileNet pre-trained on ImageNet as our convolutional base, apply data augmentation and dropout to reduce overfitting, and perform fine-tuning of the final layers to adapt the model to our task. Our final model achieves 87.9% accuracy on the validation set.

C. Model Architecture

Our model follows a Transfer Learning approach using MobileNet [3] pre-trained on ImageNet as the convolutional base. Here is the full pipeline :

1) 4317 files belonging to five classes : 3454 for training, 863 for validation (80/20).

2) Input layer: accepts 128x128x3 RGB images.

3) Data augmentation layers: RandomFlip, RandomRotation, RandomZoom.

4) Rescaling layer: normalizes pixel values from [0,255] to [0,1].

5) MobileNet base (frozen during Feature Extraction): extracts high-level feature representations from the input images.

6) Flatten layer: converts 2D feature maps to a 1D vector.

7) Dense layer (128 units, ReLU activation): learns to combine MobileNet features on task-specific flower recognition problem + ReLU activation : removes negative values, keeping only the most relevant activation.

8) Dropout (rate=0.5): randomly turn off 50% of neurons during training to prevent overfitting.

9) Output layer (5 units, Softmax activation): give a probability distribution to the five flower classes (sum=1).

The model is compiled using the Adam optimizer and sparse categorical cross-entropy loss.

II. RELATED WORKS

Several works have explored flower recognition and image classification using both traditional machine learning and deep learning approaches.

Nilsback and Zisserman [1] introduced one of the earliest flower recognition datasets (Oxford Flower 17 Dataset), it contained 17 categories of common UK flowers with 80 images per class. The image presents large variations in scale, pose, and lighting, as well as close visual similarity between certain classes. Their approach relied on handcrafted visual features — specifically a visual vocabulary built from descriptions — combined with SVMs for classification. This method demonstrated that color and texture features are highly discriminative for flower species.

Krizhevsky, Sutskever and Hinton [2] proposed AlexNet, a pioneering deep CNN with 60 million parameters trained to classify 1.3 million high-resolution images from ImageNet into 1000 classes. Their architecture consists of five convolutional layers followed by max-pooling layers and two fully connected layers with a final softmax classifier. To accelerate training they used non-saturating neurons (ReLU) and an efficient GPU implementation. Although not specific to flower recognition, AlexNet established key practices — ReLU activations, dropout regularization, and data augmentation

Howard et al. [3] introduced MobileNet, a class of lightweight CNN architecture designed for mobile and embedded vision applications. MobileNets are based on depthwise separable convolutions to reduce computation while maintaining accuracy; they also introduce two simple global hyper-parameters that efficiently trade off between latency and accuracy. Pre-trained on ImageNet, MobileNet serves as an efficient feature extractor for transfer learning in domains with limited data which is why this is the architecture that we selected for our flower recognition system.

D. Training Strategy

Training is conducted in two consecutive phases:

Feature Extraction: In the first phase, the MobileNet base is completely frozen (trainable=False). Only the Dense and Dropout layers are trained. This prevents the randomly initialized Dense layers from destroying the pre-trained ImageNet weights through large gradient updates. The model is trained for 10 epochs with a standard Adam optimizer. After this phase, the model reaches 85.7% validation accuracy.

Fine-Tuning: In the second phase, the last 20 layers of MobileNet are unfrozen while the earlier layers remain frozen. The model is re-compiled with a very low learning rate (1e-5) to perform small, careful weight updates that adapt well to MobileNet representations to our specific flower dataset without destroying the general features learned on ImageNet. The model is trained for an additional 10 epochs. After this phase, validation accuracy reaches 87.9%.

E. Results

The training curves show a consistent improvement in both training and validation accuracy across both phases. During the Feature Extraction phase, the validation accuracy quickly stabilizes around 85-86%, while the training accuracy catches up progressively. During Fine-Tunning phase, both curves converge further and reach approximately 87-88%.

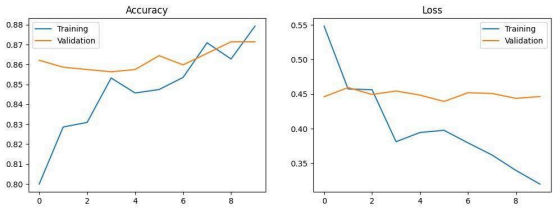


Fig. 1. Training and Validation Accuracy/Loss curves (after Fine-Tuning phase).

The confusion matrix confirms that the model performs well across all five classes. Dandelion (178/206 correct) and tulip (180/204 correct) achieve the best recall. Sunflower also shows strong performance (163/172 correct). The most confused class is rose, which is sometimes misclassified as tulip (16 errors), likely due to their similar petal shapes and color range.

III. PROPOSITION

A. Dataset

We use the Flowers Recognition dataset available on Kaggle, which contains 4,317 images organized into five categories: daisy, dandelion, rose, sunflower, and tulip. All images are in JPEG format and present significant variability in terms of scale, orientation, lighting conditions, and background complexity. The dataset is split with 80% of images used for training (approximately 3,450 images) and 20% reserved for validation (approximately 867 images). Images are resized to 128x128 pixels and normalized to the [0,1] range before being fed into the model.

B. Data Augmentation

To improve generalization and reduce overfitting given our limited dataset size, we apply on-the-fly data augmentation directly within the model pipeline. The augmentation operations include random horizontal flipping, random rotation up to 10%, and random zoom up to 10%. These transformations are applied exclusively during training and ensure that the model never sees the exact same image twice across epochs. This forces the network to learn generalizable features rather than memorizing training samples.

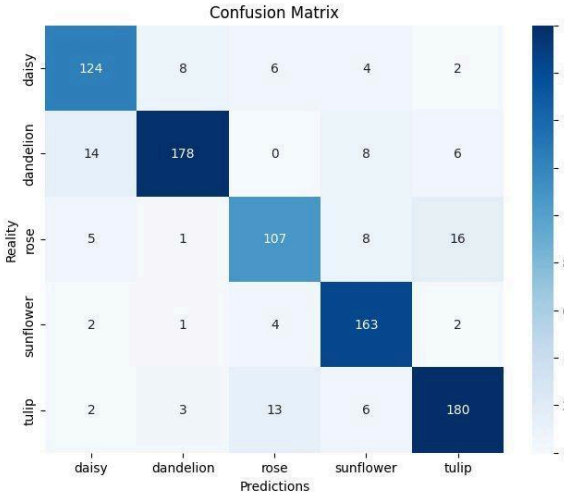


Fig. 2. Confusion Matrix on the validation set.

CONCLUSION

In this paper, we presented a flower recognition system based on Transfer Learning with MobileNet. By combining data augmentation, a frozen convolutional base for feature extraction, and fine-tuning of the deeper layers, we achieved 87.9% validation accuracy on a five-class flower dataset with 4,317 images. These results confirm that Transfer Learning is an effective strategy when working with limited labeled data.

This project allowed us to apply and consolidate the key concepts studied throughout the course: convolutional networks, overfitting prevention through dropout and data augmentation, and transfer learning with fine-tuning. Future improvements could include using a larger image resolution, testing other pre-trained architectures or extending the dataset with additional classes.

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